

Quiz 5-B

First and Last Name: _____

You have 15 minutes to complete the quiz. Calculators, notes, and other resources are **not** allowed.

1. Consider $\mathbb{R}^4$, and consider the subset W of vectors $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ where $3x_1 + 5x_3 = 0$ and $2x_2 = x_4$. In other words, $W = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \mid 3x_1 + 5x_3 = 0, 2x_2 = x_4 \right\}$. Justify that W is a subspace by showing the following:

- (a) (1 point) Show that W is nonempty.

Solution: We have $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ is in W , because $3 \cdot 0 + 5 \cdot 0 = 0$ and $2 \cdot 0 = 0$.

- (b) (3 points) Show that W is closed under addition.

Solution: For any $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$ in W , that is $3u_1 + 5u_3 = 0$, $2u_2 = u_4$ and $3v_1 + 5v_3 = 0$, $2v_2 = v_4$, we have

$$3(u_1 + v_1) + 5(u_3 + v_3) = (3u_1 + 5u_3) + (3v_1 + 5v_3) = 0 + 0 = 0,$$

and $2(u_2 + v_2) = 2u_2 + 2v_2 = u_4 + v_4$. Thus, $\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \\ u_4 + v_4 \end{bmatrix}$ is in W .

(part (c) is on the back!)

- (c) (2 points) Show that W is closed under scalar multiplication. Below is the definition of W again, so you don't have to flip back and forth:

$$W = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \mid 3x_1 + 5x_3 = 0, 2x_2 = x_4 \right\}$$

Solution: For any $\mathbf{u}$ in W and any real number c , $c \cdot \mathbf{u} = \begin{bmatrix} cu_1 \\ cu_2 \\ cu_3 \\ cu_4 \end{bmatrix}$, we have

$$3(cu_1) + 5(cu_3) = c(3u_1 + 5u_3) = 0, \quad 2(cu_2) = c(2u_2) = cu_4.$$

Thus, $c\mathbf{u}$ is in W .